

object $\rightarrow$ keys/value or method (function)
order fixed nhi hota like array

ob = { }

Student = {
 name: 'Adi',
 class: 5,
 Subject: ['H', 'E', 'M'],
 Rollno: 49
}

object

$\rightarrow$ student.name
 student["name"]

$\rightarrow$ object.values(student)
 $\rightarrow$ object.keys(student)
 $\rightarrow$ object.entries(student)

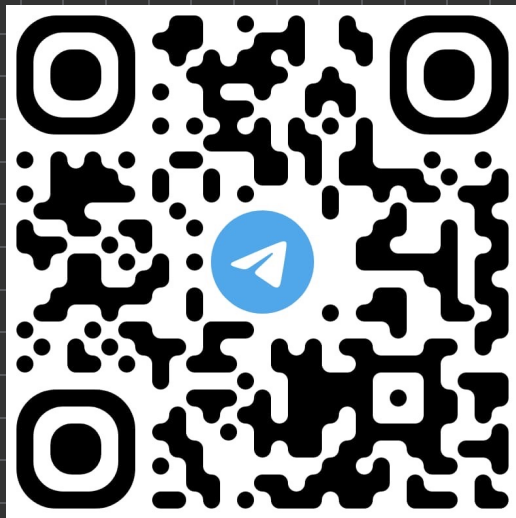
} array

for in $\rightarrow$ object (keys) array (index)
 for of $\rightarrow$ array

rest $\rightarrow$ pack $\rightarrow$ array destructuring
 spread $\rightarrow$ array $\rightarrow$ unpack

arr = [1, 2]
 let [n1, n2] = [1, 2]
 = arr

Telegram



Discord



object destructuring

Array, object → update → let, const

→ splice, slice, find, indexOf, flat

method

→ Map, filter & Reduce

updated key name

```
// how to rename key  
let {subjects : vishay , ...variable } = student  
// subjects -> vishay  
// let vishay = subjects  
console.log(vishay);
```

```
let india = "country"
```

```
let obj2 = {
```

```
  address : india,
```

```
  adharCard : 5223452345543
```

```
}
```

variable



```
let obj1 = {  
  name: "nishant",  
  phone: 7825763245  
}
```

```
let obj2 = {  
  address: "india",  
  adharCard: 5223452345543,  
  name : "yash"  
}
```

```
let obj3 = {...obj1 , ...obj2}  
console.log(obj3);
```

yash

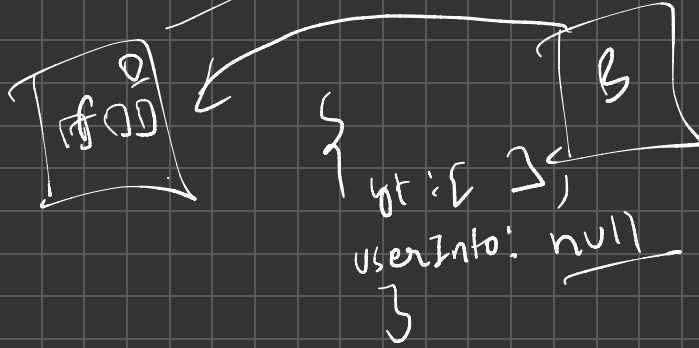
{ name, phone, add, adh, }

name
yash
nishant

{ name, phone, add, adh }

without
Account

youtube



{

userInfo: " " }

optional chaining

?.

address: {
 street
 pincode
}

shift $\rightarrow$ start delete

unshift $\rightarrow$ start add

push $\rightarrow$ last add

pop $\rightarrow$ last delete

arr = [1, 2, 3, 4, 5]

splice → Add, delete | Replace

.splice(start, deleteCount)

↑
start
index

Jagah se
delete krna
ho

↓
kitna delete
krna hai start se
(start value include hai)

delete

add

replace

(delete + add)

slice



Array ko trim krna

`.slice( start index , end index )`
included excluded

.indexof(^{value})

if present return index

else -1 (value not present in the array)

• find(0 → 23)

↳ kisi bhi element ko find
karta hai based on
your condition

first element

that satisfies
condition

Mutability

changable

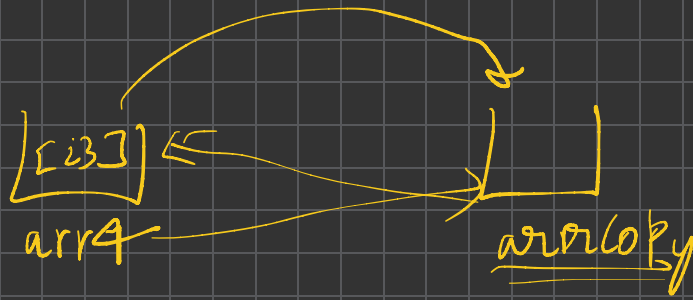
original variable
ko bhi change
kar deta hai

splice	shift
pop	push
unshift	

Immutability

nonchangable
(return new array)

slice



value copy X
reference copy

forEach

Map, filter & Reduce → Immutable
↓ array methods

Map → Methods
↓
array input ~ length
↓
return new array of n length

iterate

```
let originalPrices = [463 , 654 , 2346]
```

```
let discountedPrices = [] ✓
```

```
for(value of originalPrices){  
  discountedPrices.push(value * 0.9) // 10% discount  
}
```

```
console.log(originalPrices);  
console.log(discountedPrices);
```

```
let originalPrices = [463 , 654 , 2346]  
let discountedPrices = []  
  
for(value of originalPrices){  
  let discount = value * 10 / 100  
  discountedPrices.push(value - discount) // 10% discount  
}  
  
console.log(originalPrices);  
console.log(discountedPrices);
```

100% → 1
10% → 0.1
1.0 - 0.1
0.9

→ 90%

1000 → 100%
-10%]

90%

900

90% of 400

400 * 90 / 100

dfs

```
let originalPrices = [463 , 654 , 2346]
```

```
let discountedPrices = []
```

```
for(value of originalPrices){
```

```
  let discount = value * 10 / 100  
  discountedPrices.push(value - discount) // 10% discount
```

```
  discountedPrices.push(value * 0.9) // 10% discount  
}
```

```
console.log(originalPrices);
```

```
console.log(discountedPrices);
```

discounted

[416, ~~590~~, 2300]

```
let students = [  
  {  
    name: "Ayaan",  
    marks: 56,  
  },  
  {  
    name: "Mansi",  
    marks: 46,  
  },  
  {  
    name: "Debadrita",  
    marks: 33,  
  },  
  {  
    name: "Shivan",  
    marks: 30,  
  },  
]
```

→ array of objects

[{ }, { }, { }, { }]

array of videos

[{ }, { }, { }, { }]

title
number |

```
let boostedMarks = students.map((student) => {  
  return {...student, marks : student.marks + 10}  
})
```

↑
spread

{ name, marks, marks }

[] => { }

{ }

[{ }, { }, { }]

filter $\rightarrow$

(
array input $\rightarrow$ N length
return new array $0 - N$ length

chaining

method1().method2().method3()

```
[
  { name: 'Ayaan', marks: 66 },
  { name: 'Mansi', marks: 56 },
  { name: 'Debadrita', marks: 43 },
  { name: 'Shivan', marks: 40 },
  { name: 'Alauddin', marks: 38 }
]
```

chaining

array

```
[ { name: 'Shivan', marks: 30 }, { name: 'Alauddin', marks: 28 } ]
```

filter result

['Shivan', 'Alauddin']

```
const failedStudents = students.filter((student) => student.marks < 33).map((student) => student.name)
```

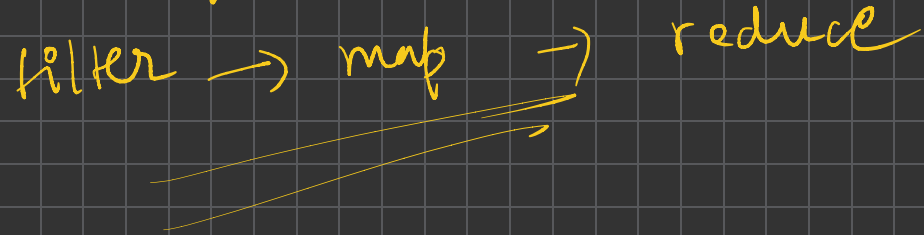
```
//filter ak use krke agar hmko marks hi dikhane h name nhi to sir
```

```
// const failedStudentsName = failedStudents.map((student) => student.name) ✓
```

```
console.log(failedStudents);
```

chaining

filter → map → reduce



```
graph LR; filter --> map; map --> reduce
```

Reduce

(array n length

) return single value (num, boolean, object
array)

Method	Purpose	Callback Return Value	Output Array Length
<code>map()</code>	Transforms each element	The new transformed value	Same length as original
<code>filter()</code>	Filters out unwanted elements	<code>true</code> to keep, <code>false</code> to discard	Usually shorter than original
<code>reduce()</code>	Condenses array to one value	The updated accumulator	Single value (any data type)

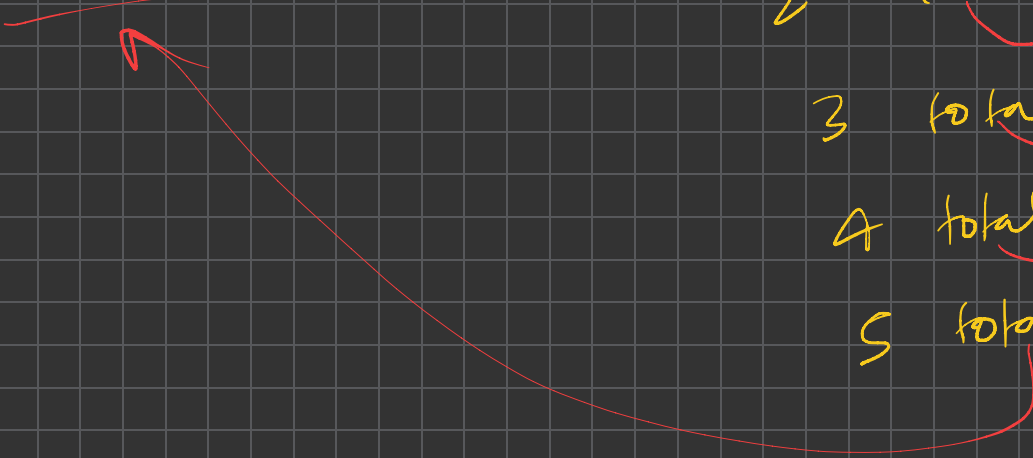
⁺ ⁺ ⁺ ⁺

```
let marks = [56, 24, 62, 73, 78]

let totalMarks = 0

marks.forEach((mark) => totalMarks = totalMarks + mark)

console.log(totalMarks);|
```



- 1 $total = total + mark$
 $0 + 56$
 $total = 56$
- 2 $total = 56 + 24$
- 3 $total = total + 62$
- 4 $total = total + 73$
- 5 $total = total + 78$

→ totalM = 0
[1, 2, 3, 4]
• reduce (callback-fn)
↓ accumulator

```
const totalMarks = marks.reduce((accumulator, currentValue) => {}, 0)
```

↑ initial Value

```
let marks = [56, 24, 62, 73, 78]
```

```
const totalMarks = marks.reduce((accumulator, currentValue) => {  
  accumulator = accumulator + currentValue  
  return accumulator;  
}, 0)
```

0, 56, 80

acc	cur	=	acc + cur
0	56		<u>56</u>

acc	cur	=	acc + cur
56	24		80

acc	cur	=	acc + cur
80	62		

```
const attendance = ["present", "present", "absent", "present", "absent"]
```

```
// -> { present : 3 , absent : 2 }
```

```
let obj = {}
```

```
attendance.forEach((value) => {  
  if (obj[value]) {  
    obj[value] = obj[value] + 1  
  } else {  
    obj[value] = 1  
  }  
})
```

```
console.log(obj);
```

$obj = \{ \}$

↗ false

↗ falsy

"present"

↗ undefined

$obj["present"]$

→

$obj["present"] = 1$

$obj["present"] = 2 + 1$

$obj = \{ \text{present: } 1, \text{absent: } 1 \}$

```
const obj = attendance.reduce((acc, value) => {  
  acc[value] = (acc[value] || 0) + 1;  
  return acc  
}, {})
```

↓
0
undefined
false

short circuit
in logical operator

} }
present